WORKSHEET 9.3

Calculator Strategy

Unit 9: Exam Strategy and Free-Response Types

Key ideas

- The calculator is for four jobs: graphing in a window, solving an equation, a derivative at a point, and a definite integral.
- Write the mathematics, not the keystrokes: the integral or equation in standard notation, then $\approx$ and the value.
- Store intermediate values. Round only the final answer, to three decimal places.
- Radian mode, always.

Worked examples

Example 1 · Solving

Solve $x^3 + x = 5$.

Solution. Graph $y = x^3 + x - 5$ and find its zero.

$x \approx 1.516$.

Example 2 · A derivative at a point

Find $f'(2)$ for $f(x) = \sqrt{x}\,\sin x$.

Solution. Use the numerical derivative at $x = 2$.

$f'(2) \approx -0.267$.

Example 3 · A definite integral

Evaluate $\displaystyle\int_0^2 \sqrt{1 + x^3}\,dx$.

Solution. Use the numerical integral.

≈ 3.241.

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Practice

A calculator is required on this page. Give decimal answers to three places.

CORE

1. Evaluate $\int_0^1 e^{x^2}\,dx$.

2. Solve $\ln x + x = 3$.

3. Find $f'(0.5)$ for $f(x) = \tan(x^2)$.

4. A calculator displays $7.34846\ldots$ Which of 7.3, 7.34, 7.348 and 7.349 are acceptable final answers?

EXAM LEVEL

5. Find the area between $y = \sin(x^2)$ and the x-axis for $0 \le x \le \sqrt{\pi}$.

6. A particle has $v(t) = \ln(t+1) - 1$ for $0 \le t \le 4$. Find the total distance it travels. Write the integral first.

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7. Find the average value of $f(x) = \sqrt{x}\cos x$ on $[0, 3]$.

8. A rate is modelled by $R(t) = 10te^{-0.4t}$ for $t \geq 0$. Find both times at which $R(t) = 6$.

Question 9 · Free-response practice

Rain falls at the rate $r(t) = 5\sqrt{t}\,e^{-0.3t}$ millimetres per hour for $0 \leq t \leq 8$, where t is in hours.

(a) Find the total rainfall for $0 \leq t \leq 8$.

(b) Find $r'(3)$, and interpret it in context with units.

(c) Find the interval of time during which rain is falling at more than 3 millimetres per hour.

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Worked solutions

1. ≈ 1.463.
2. $x \approx 2.208$.
3. $f'(0.5) \approx 1.065$.
4. 7.348 (rounded or truncated). The value 7.349 is wrong, and the others have too few places.
5. $\displaystyle\int_0^{\sqrt{\pi}} \sin\left(x^2\right)\, dx \approx 0.895$.
6. $\displaystyle\int_0^4 |\ln(t+1) - 1|\, dt \approx 1.484$.
7. $\displaystyle\frac{1}{3}\int_0^3 \sqrt{x}\cos x\, dx \approx -0.216$.
8. $t \approx 0.839$ and $t \approx 5.571$.
9. (a) $\displaystyle\int_0^8 r(t)\, dt \approx 21.923$ millimetres.
   (b) $r'(3) \approx -0.469$. At $t = 3$ hours the rate of rainfall is decreasing at about 0.469 millimetres per hour per hour.
   (c) $r(t) = 3$ at $t \approx 0.480$ and $t \approx 4.023$, and $r(t) > 3$ between them: $0.480 < t < 4.023$.

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